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Learning an attention-aware parallel sharing network for facial attribute recognition☆

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ABSTRACT

Existing multi-task learning based facial attribute recognition (FAR) methods usually employ the serial sharing network, where the high-level global features are used for attribute prediction. However, the shared low-level features with valuable spatial information are not well exploited for multiple tasks. This paper proposes a novel Attention-aware Parallel Sharing network termed APS for effective FAR. To make full use of the shared low-level features, the task-specific sub-networks can adaptively extract important features from each block of the shared sub-network. Furthermore, an effective attention mechanism with multi-feature soft-alignment modules is employed to evaluate the compatibility of the local and global features from the different network levels for discriminating attributes. In addition, an adaptive Focal loss penalty scheme is developed to automatically assign weights to handle the problems of class imbalance and hard example mining for FAR. Experiments demonstrate that the proposed method achieves better performance than the state-of-the-art FAR methods.

1. Introduction

Facial attributes, such as age, gender and race, are used to describe the specific characteristics of a human face and have widespread applications in image search [1], face verification [2,3], social media [4,5], etc. Recent studies have shown that facial attribute recognition (FAR) is one of the most popular research topics in computer vision and pattern recognition. However, learning a robust model for FAR is still a challenging problem due to the influence of pose variations, illumination changes, background clutters and so on.

During the past decades, significant progresses have been made in FAR [4,6–8]. In contrast to the traditional feature extraction methods [5,9] using hand-crafted design features such as SIFT [5] and HOG [9], the deep learning technology [6,10] can automatically learn features from a large amount of data. With the great success of deep learning in various computer vision tasks [11–13], there emerges an increasing number of deep learning based FAR methods [7,8,14]. According to the facial region size of feature extraction, deep learning based FAR methods are roughly divided into part-based methods [15–17] and holistic methods [7,18,19]. The part-based methods extract features from different facial parts corresponding to facial attributes. So the part-based methods rely on the accuracy of face location and alignment that may fail for complex facial variations. On the other hand, holistic methods obtain features from the entire face image, where an additional step of locating facial parts is not required. Recently, due to the application of multi-task learning (MTL) [14,20–22], the performance of FAR has been significantly improved by exploiting the intrinsic relationships among attributes. In this paper, we mainly focus on MTL-based holistic FAR methods [7,18,19].

The traditional MTL-based holistic FAR methods usually employ the serial sharing network, as shown in Fig. 1(a), which shares the features of all attributes at low-level layers while adopting multiple branches to predict attributes at high-level ones. However, only the high-level abstract features from the end of each branch are used for the final attribute prediction. Consequently, the low-level shared features with valuable spatial information and the detailed facial features may not be fully explored to improve the overall performance. Moreover, a lot of distractive information is learned from the irrelevant facial regions, which might degrade the recognition performance for the holistic FAR methods. Besides, the training set of FAR often suffers from the class imbalance problem of the attributes (e.g., the positive samples of Bald, Mustache and Gray Hair attributes are extremely rare), which may result in decreasing the recognition rate.

To address the above problems, we propose a novel Attention-aware Parallel Sharing (APS) network for FAR. In the parallel sharing

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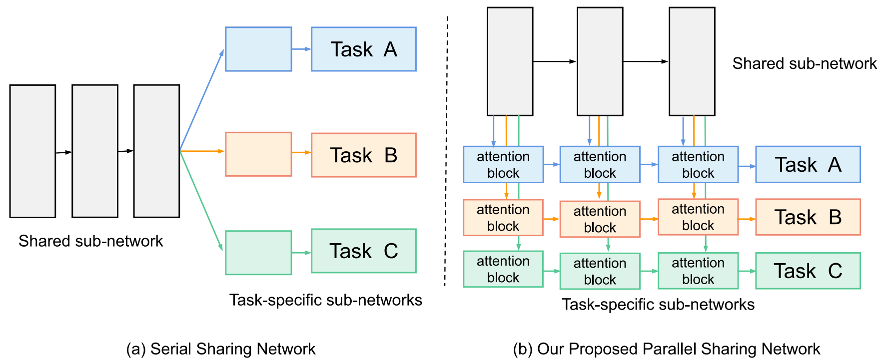

Fig. 1. Comparison between two kinds of deep networks based on multi-task learning. (a) For the serial sharing network, each task-specific sub-network is learned only behind the shared sub-network. (b) For the parallel sharing network, each task-specific sub-network can be learned from each convolutional block of the shared sub-network.

network, each task-specific sub-network obtains the attention-aware features from each convolutional block of the shared sub-network, rather than only from the end of the shared sub-network in the traditional serial sharing network [19,23], as illustrated in Fig. 1(b). The proposed network can automatically adjust the importance of different blocks of the shared sub-network for different tasks. This flexibility enables much more powerful feature representations to be learned for the shared features across tasks, while still allowing for the specific features to be maximally preserved for each individual task. Furthermore, an attention mechanism with the multi-feature soft-alignment modules is designed to evaluate the compatibility of features from different network levels in both the shared sub-network and the task-specific sub-networks. This design is to allow important features of different network levels to participate in the final attribute classification, rather than only the high-level features at the end of the network. In addition, we also develop an adaptive Focal loss penalty scheme, which can mine the hard samples and alleviate the class imbalance problem. Finally, the experimental results on the public datasets show that the proposed method achieves better performance than state-of-the-art FAR methods.

The main contributions of our work are summarized as follows:

- A novel attention-aware parallel sharing network for FAR is proposed to improve the recognition performance of the task-specific sub-networks by making full use of the attention-aware features from different blocks of the shared sub-network in the multi-task learning framework.
- An effective attention mechanism with multi-feature soft-alignment modules is employed to evaluate the compatibility of the features from the different network levels in the shared and task-specific sub-networks. The multi-feature soft-alignment module can capture important features from the different network levels, and these features will participate in the final attribute classification to further improve the performance of FAR.
- We also develop an adaptive Focal loss function to handle the problems of class imbalance and hard example mining for attribute recognition. An adaptive Focal loss penalty scheme is used to smoothly adjust the imbalance degree, which is effective to improve the recognition accuracy.
- Extensive experimental results on the challenging publicly available datasets illustrate the effectiveness of the proposed APS method for FAR.

The remainder of this paper is organized as follows. In Section 2, we review related works. In Section 3, we introduce the proposed APS method in detail. In Section 4, extensive experiments and results on the challenging CelebA and LFWA datasets are reported. Finally, we conclude this work in Section 5.

2. Related work

The proposed method is closely related to deep learning based FAR, multi-task learning and attention mechanism. The relevant literature is reviewed in the following subsections.

2.1. Facial attribute recognition

Different from the multi-class classification problems, e.g., face recognition [24], age recognition [25], and micro-expression recognition [26], facial attribute recognition (FAR) is a multi-label classification problem, which requires to predict multiple facial attribute labels. With the development of deep learning, state-of-the-art FAR methods have made remarkable progress. According to the different patterns of model construction, deep learning based FAR methods are generally classified into two categories, i.e., part-based methods [5,6,15–17] and holistic methods [27–31].

On the one hand, the part-based methods [2,6,15–17] first locate the attribute-related facial regions and then extract features from each region. For example, Kumar et al. [2] proposed to extract the low-level features from the manually defined facial regions, and then the Support Vector Machines (SVM) with radial basis function kernels is trained for FAR. Zhang et al. [16] employed a PANDA model which trains a CNN on the corresponding facial region for each poselet, and then combines the features from all these poselets for further predicting attributes. Liu et al. [6] developed a cascade of two CNNs, i.e., the Localization Networks (LNets) to locate the faces and an Attribute Network (ANet) to extract facial attribute features, which are used to learn a SVM classifier for each facial attribute. Ouafi et al. [32] proposed a two-stages-estimator (TSE) for age estimation based on facial demographic attributes. The first stage of TSE divides the input face images into the different gender-race groups, and then the second stage learns the gender-race groups of each classification to estimate ages. In summary, the part-based FAR methods can avoid the negative influence of irrelevant region information, but the performance of these attribute recognition methods might decline if the region localization fails.

On the other hand, in order to avoid the inaccurate region localization, the holistic FAR methods [27–31] extract features from the entire image without relying on any localization mechanism. Zhong et al. [27] proposed to use the mid-level CNN features as an alternative to the high-level features for attribute prediction and verified the importance of mid-level CNN features for attribute recognition. Sharma et al. [28] designed an efficient deep neural network, named Slim-CNN, which introduced a computationally efficient Slim module. The Slim module can be viewed as a micro-architecture CNN composed of depth-wise separable convolutions and point-wise convolutions. Zhuang et al. [29] proposed a multi-label learning based deep transfer neural network, which reduces the dependence on fully labeled training data, especially when some attributes lack the label information. Bhattarai et al. [30] generated word embedding of categorical labels for images to minimize the categorical loss. Yang et al. [31] developed a feature separation model exchange-GAN, which can separate expression-related features and expression-independent features, for obtaining the high expression recognition accuracy. Hassanat et al. [25] proposed a deep learning network DeepVeil based on VGG-19 to recognize face, age and expression of veiled-person faces. In short, the holistic FAR methods assume that each facial region should be equally considered, and they explicitly exploit the correlations between the attributes. However, the final attribute prediction of most existing methods just rely on the abstract features from the high-level layers of the network, thus ignoring the fact that the low-level layers might contain the useful spatial information for discriminating attributes.

2.2. Multi-task learning based FAR

Multi-task learning (MTL) [33] employs the useful information between multiple tasks to improve their respective performance. MTL has found success in various computer vision tasks [20–22]. In recent years, state-of-the-art FAR methods employ multi-task neural networks to predict the attributes and have shown remarkable improvements in performance. MTL generally shares feature information at low-level layers for all the tasks and adopts multiple branches to predict attributes at high-level ones. For instance, Rudd et al. [18] introduced a mixed objective optimization network (MOON) based on MTL, and designed a domain-adapted multi-task loss function to solve the class imbalance problem. Sun et al. [34] proposed a general-to-specific learning strategy to extract features, and the learning strategy contains three steps, i.e., joint learning, task-aware learning and attribute-aware face cropping. By step-to-step learning, this method can improve the attribute prediction in the wild. Wang et al. [35] designed a recognizer and a discriminator for multi-task face analyses (i.e., facial landmark detection, head pose and gender recognition, and FAR). Based on multi-task learning and adversarial learning, the performance of each face-related task can be boosted in this method. Shu et al. [36] proposed to use multiple auxiliary tasks (i.e., patch rotation, patch segmentation and patch classification) to learn the spatial-semantic relationship information from unlabeled facial data, and then transfer the spatial-semantic relationship information to the target attribute recognition task, thus improving the performance of FAR with only limited labeled data.

The recent MTL methods [7,23,37] try to group attributes for different tasks. For example, Hand et al. [23] proposed a multi-task deep network to fully utilize the implicit and explicit relations among attributes, and built an auxiliary layer that learns attribute relations at the score level to improve the final classification performance of each attribute. Han et al. [7] proposed to consider the correlation and heterogeneity among attributes and adopted a deep multi-task learning (DMTL) network for predicting attributes. In [7], the attributes are divided into the four different heterogeneous attribute categories, and then this network learns the shared features from all attributes and the task-specific features from four heterogeneous attribute groups, respectively. Different from manual attribute grouping, Fanhe et al. [37] developed an automatic grouping scheme based on training data itself, by calculating spectral clustering and the correlation of attributes to divide the attributes into three groups with strong or weak label for multi-task learning. Mao et al. [19] developed a deep multi-task multi-label CNN (DMM-CNN), which divides the attributes into objective and subjective attribute groups to consider the different learning complexity of facial attributes and joints the facial landmark detection to improve the performance of FAR. Cao et al. [38] proposed a partially shared multi-task convolutional neural network with local constraint (PS-MCNN-LC) for FAR, which designs partially shared structures to learn the feature representations of the task-specific networks (TSNets) and the shared network (SNet). But PS-MCNN-LC connects TSNets and SNet only by a simple feature concatenation, and it needs to laboriously choose the numbers of channels of TSNets and SNet by experiments. In addition, it defines a loss function LCLoss which requires the additional identity information.

However, there exist some limitations on deep multi-task learning based FAR methods [7,18,23,34,37]. The general strategy of the traditional multi-task learning methods is to use the serial sharing network, where the useful low-level features are not be fully explored for all the tasks. Moreover, it is difficult to determine the split point of the shared sub-network for multiple tasks to obtain the optimal recognition performance. Different from the existing methods, we propose a novel parallel sharing network, which contains multiple attention blocks to fuse the feature information of different levels of both the shared sub-network and each task-specific sub-network, in order to make full use of shared information for multi-task learning.

2.3. Attention mechanism

The attention mechanism is widely used in various classification tasks [39–42] to imitate human vision by focusing on the target area that needs to be paid attention to. It devotes more attention to the informative image region, while suppressing the irrelevant and potentially confusing information in other regions. Wang et al. [39] proposed a residual attention network constructed by stacking encoder–decoder attention modules to generate attention-aware features. This attention mechanism can achieve better performance even in very deep networks. Hu et al. [40] proposed a Squeeze-and-Excitation (SE) block to enhance the representational power of the network. The SE block can be inserted into many existing classification networks, and moreover, it has the advantages of simple design, less parameter settings, and less calculations. Woo et al. [41] proposed a simple yet effective attention module, named Convolutional Block Attention Module (CBAM), which contains two separate dimensional attention sub-modules, i.e., channel-wise attention and spatial-wise attention. Compared with the attention mechanism only using channel-wise attention in the SE block [40], the CBAM can achieve better results. Yang et al. [42] designed a novel image super-resolution (SR) network with multiple attention mechanism. In order to improve the ability of feature expression, the SR network uses a channel-wise attention mechanism to consider the relationship between feature channels, and employs a spatial-wise attention mechanism to learn the relationship between pixels in the feature map.

Despite their success, these attention modules neither make full use of the global and local features of images, nor consider the relationships between the shared sub-network and the task-specific sub-network. To overcome these shortcomings, we introduce a novel attention mechanism with the multi-feature soft-alignment modules to predict attributes by combining the features of different network levels from both the shared sub-network and the task-specific sub-networks.

3. Proposed method

In this section, an overview of the network architecture of the proposed APS method is first introduced, and the main components of the proposed method are then described in detail.

3.1. Overview

The network architecture of the proposed APS method is shown in Fig. 2. In the APS method, we design an attention-aware parallel sharing network, which consists of a shared sub-network and two task-specific sub-networks. The shared sub-network contains multiple convolution layers based on VGG-16 and extracts shared features of an input image. According to the different facial areas of interest, we first divide all the facial attributes into partial attributes and general attributes based on [43]. Such a design is based on the different characteristics of partial attributes and general attributes. To be specific,

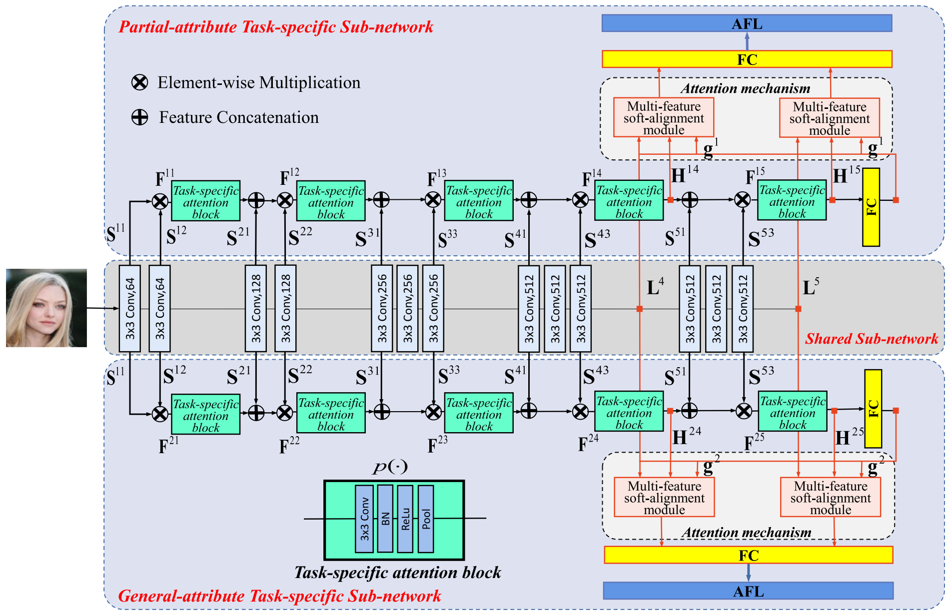

Fig. 2. The overall framework of the proposed APS method. VGG-16 is used to extract the shared features, and the two task-specific sub-networks are respectively trained for partial attributes and general attributes. The two multi-feature soft-alignment modules are introduced at the different levels of the shared and task-specific sub-networks. Note that the task-specific attention block is shown separately for clarity.

the partial attributes are closely associated with a certain part of the face (such as Eyeglasses and Bags_Under_Eyes). General attributes (such as Attractive and Young) are generally not limited to a certain part of the face but focus on the whole face. Therefore, we further design the partial-attribute and general-attribute task-specific sub-networks for two different tasks to recognize partial and general attributes, respectively. More specifically, the different layers of each task-specific sub-network are connected with the shared sub-network through a parallel sharing network framework (see Section 3.2). Moreover, an attention mechanism is employed for each task-specific sub-network, where each multi-feature soft-alignment module in the attention mechanism takes the feature maps from the different levels of both the shared sub-network and the task-specific sub-networks. Then, these features are combined with the linear classification layer at the end of the task-specific sub-network for more accurate attribute classification (see Section 3.3). Finally, the APS method adopts an effective joint loss function to improve the recognition rate with an adaptive Focal loss penalty scheme (see Section 3.4).

3.2. Parallel sharing network

To make full use of the shared network features that contain valuable spatial information and detailed facial features, a novel parallel sharing network is designed for multi-task learning of FAR, where a shared sub-network is constructed for shared feature learning and the partial-attribute and general-attribute task-specific sub-networks are exploited for learning the features of the two respective attribute groups. In the parallel sharing network, the shared sub-network can interact with each task-specific sub-network to promote the information flow from the shared layers to the task-specific layers.

As illustrated in Fig. 2, the partial-attribute and general-attribute task-specific sub-networks respectively contain a set of task-specific attention blocks, each of which obtains the additional inputs from the previous layers of the shared sub-network and then transfers the task-specific features to the next layer. We denote the shared features as $\mathbf{S}^{bl}$ from the lth layer of the bth block in the shared sub-network, and then the input features $\mathbf{F}^{tb}$ of the bth block in the tth task-specific sub-network can be formulated as follows:

$$\mathbf{F}^{tb} = \begin{cases} \mathbf{S}^{b1} \otimes \mathbf{S}^{b2}, & b = 1 \\ \left[p\left(\mathbf{F}^{t(b-1)}\right), \mathbf{S}^{b1} \right] \otimes \mathbf{S}^{b2}, & b = 2 \\ \left[p\left(\mathbf{F}^{t(b-1)}\right), \mathbf{S}^{b1} \right] \otimes \mathbf{S}^{b3}, & b = 3,4,5 \end{cases}, \tag{1}$$

where $\otimes$ denotes the element-wise multiplication, $[\cdot,\cdot]$ denotes the feature concatenation, and $p(\cdot)$ represents a sequence of operations of the task-specific attention block for each task, i.e., Conv-BN-ReLU-Pool. As shown in Fig. 2, the first block of the task-specific sub-network only takes the features in the shared sub-network as the input, where $\mathbf{S}^{b1}$ and $\mathbf{S}^{b2}$ are the first and second feature layers of the first shared sub-network block. But for the subsequent sub-network blocks (i.e., $b = 2, \ldots, 5$), the shared feature $\mathbf{S}^{b1}$ and the task-specific feature $p\left(\mathbf{F}^{t(b-1)}\right)$ from the previous attention block are concatenated, and then we perform an element-wise multiplication between this concatenated feature and the shared feature from the last layer of the current block. This design enables the task-specific sub-networks to leverage the feature information from the shared sub-network, thereby avoiding the disappearance of the shared features as the network layer deepens.

3.3. Multi-feature soft-alignment module

In order to exploit the features of the different network levels to participate in the final classification of FAR, we design an attention mechanism with multiple multi-feature soft-alignment modules. This module can pay attention to the correlation (i.e., compatibility in this paper) between global features and local features. The local features are extracted from the intermediate layers of both the shared sub-network and each task-specific sub-network, and the global feature is extracted from the first FC layer at the end of the network. Based on the notion of compatibility [44,45], the attention value of each multi-feature soft-alignment module is given by calculating a compatibility score between the local features and the global feature, where the local and global features are used together for the attribute classification. The attention mechanism employed in APS contains a set of multi-feature soft-alignment modules, as shown in Fig. 2.

For the tth task-specific sub-network, we convert the feature matrix $\mathbf{H}^{tb} = p\left(\mathbf{F}^{t(b-1)}\right)$ outputted from the bth block to be a set of n column vectors, i.e., $\mathbf{H}^{tb} = \left\{\mathbf{h}_1^{tb}, \mathbf{h}_2^{tb}, \ldots, \mathbf{h}_n^{tb}\right\}$, where $\mathbf{h}_i^{tb}$ is the ith vertical component of $\mathbf{H}^{tb}$. To be specific, $\mathbf{H}^{tb}$ that is a $C \times H \times W$ feature map, is reshaped to a $C \times (H \times W)$ matrix, where each $\mathbf{h}_i^{tb}$ is a column vector corresponding to the C-dimensional vector for a single spatial location, and n is the number of spatial locations for the bth block. The global feature $\mathbf{g}^t$ is the output of the first FC layer after a series of convolutions of the tth task-specific sub-network. Similarly, for the shared sub-network, the set of feature vectors $\mathbf{L}^b = \{\mathbf{l}_1^b, \mathbf{l}_2^b, \dots, \mathbf{l}_n^b\}$, can be extracted from the bth shared feature block. Here, $\mathbf{l}_i^b$ is the ith component of all n feature vectors of $\mathbf{L}^b$. Inspired by [44,45], we use the compatibility score to combine the local features from both the task-specific sub-networks and the shared sub-network with the global feature for each multi-feature soft-alignment module. The proposed method can pay more attention to those local features that are beneficial to the linear classification and suppress the irrelevant information, and thus the compatibility between the local and global features plays the role of attention.

Specifically, for the bth block of the tth task-specific sub-network, we denote the set of compatibility scores as $C\left(\mathbf{h}_i^{tb}, \mathbf{g}^t\right) = \left\{c_1^{tb}, c_2^{tb}, \dots, c_n^{tb}\right\}$. A compatibility score c_i^{tb} of the tth task-specific sub-network is defined as

$$c_i^{tb} = \left\langle \mathbf{u}, \mathbf{h}_i^{tb} + \mathbf{g}^t \right\rangle, \ i \in \{1, 2, \dots, n\} \ . \tag{2}$$

Here, it outputs a scalar compatibility score between two input vectors with the same dimension. Firstly, the local feature $\mathbf{h}_i^{tb}$ and the global feature $\mathbf{g}^t$ are connected by a simplified addition operation, without loss of generality. We then learn a single fully-connected mapping (i.e., the fully-connected layer with the weight vector $\mathbf{u}$), and the obtained features are finally mapped to the compatibility score c_i^{tb}.

Similarly, for each block b of the shared sub-network, we denote the set of compatibility scores expressed as $C\left(\mathbf{l}_i^b, \mathbf{g}^t\right) = \left\{r_1^b, r_2^b, \dots, r_n^b\right\}$. A compatibility score r_i^b of the shared sub-network is expressed as

$$r_i^b = \left\langle \mathbf{w}, \mathbf{l}_i^b + \mathbf{g}^t \right\rangle, \ i \in \{1, 2, \dots, n\} \ , \tag{3}$$

where $\mathbf{w}$ is the weight vector of the fully-connected layer. Thereby, by combining the compatibility score of the tth task-specific sub-network with that of the shared sub-network, the total compatibility score is as follows:

$$\hat{c}_i^{tb} = \tau c_i^{tb} + (1 - \tau) r_i^b \ , i \in \{1, 2, \dots, n\} , \tag{4}$$

where τ is a hyperparameter for weighting compatibility scores of different sub-networks. After the total compatibility score of the tth task-specific sub-network is calculated, a softmax function is used to normalize it:

$$a_i^{tb} = \frac{\exp\left(\hat{c}_i^{tb}\right)}{\sum_j^n \exp\left(\hat{c}_j^{tb}\right)} \ , i \in \{1, 2, \dots, n\} \ . \tag{5}$$

Therefore, the total compatibility scores of all the n feature vectors after normalization can be concatenated as $\mathbf{A}^{tb} = \left\{a_1^{tb}, a_2^{tb}, \dots, a_n^{tb}\right\}$ for the b-the block of the tth task. The new feature vector obtained from the multi-feature soft-alignment module corresponding to the bth block of the tth task-specific sub-network is as follows:

$$\mathbf{G}^{tb} = \sum_{i=1}^{n} a_i^{tb} \cdot \mathbf{h}_i^{tb} \ . \tag{6}$$

Here, the new global descriptor $\mathbf{G}^{tb}$ is obtained by the element-wise weighted averaging between the local feature $\mathbf{h}_i^{tb}$ of the tth task-specific sub-network and the normalized compatibility score a_i^{tb}.

In our work, the attention value of each multi-feature soft-alignment module is calculated based on the compatibility score between the local features and the global feature outputted from the first FC layer, and thus the compatibility score should be higher when the local features are from the deeper network levels. Besides, the features at the different network depths contain the different semantic information. In order to promote the diversity and complementarity of the attention-weighted features, we use the two multi-feature soft-alignment modules at the relatively deeper layers of each task-specific sub-network. Therefore, we use the global descriptors of the two multi-feature soft-alignment modules corresponding to the last two blocks (i.e., b = 4, 5) for each task-specific sub-network, and these two global descriptors can be concatenated as

$$\mathbf{G}^t = \left[\eta \mathbf{G}^{t4}, (1 - \eta) \mathbf{G}^{t5}\right] \ . \tag{7}$$

Here, η is a tuning hyperparameter. As a result, the output of the attention mechanism denoted as $\mathbf{G}^t$ is fed into the linear classifier for generating the prediction scores of the corresponding attributes, which can combine the local and global features of the different sub-networks to make a classification decision. The global feature is the output of the first FC layer at the end of the network, so the compatibility score should be higher if the local features contain the more discriminative classification information. Such a design makes the network focus on the image regions beneficial to classification, and reduces the influence of the irrelevant and easily confused regions.

3.4. Adaptive focal loss penalty scheme

We further employ a novel adaptive Focal loss penalty scheme to address the problems of class imbalance and hard example mining of facial attributes. Given a training dataset with N facial images, and their corresponding labels of M attributes. Based on [18,19], we first introduce the mean square error (MSE) loss for each attribute during training, which is expressed as

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{M} \left(\hat{y}_{ij} - y_{ij}\right)^2 \ . \tag{8}$$

Here, $y_{ij} \in \{0, 1\}$ is the ground truth, and $\hat{y}_{ij}$ is the estimated probability of the jth attribute for the ith image, which is outputted from the final FC layer. However, during this process, the vast number of easily classified samples can overwhelm training, thus leading to low accuracy. To deal with the above problem, we firstly define a factor $\left(1 - \hat{y}_{ij}\right)^\gamma$ to provide the weights of hard examples for the estimated probability $\hat{y}_{ij}$. In this way, it can reduce the weights of easy examples and thus focus training on hard examples.

In addition, the traditional Focal loss [46] uses a fixed parameter α to handle the class imbalance problem. However, the fixed parameter is difficult to adapt to the training of the samples of different batches. Therefore, in order to further solve the class imbalance problem, we introduce an adaptive parameter to balance the different classes by calculating the ratio of positive samples in each batch of training data. Thus an adaptive Focal loss penalty term is defined as

$$L_{AF} = \begin{cases} \left(1 - q_j\right) \cdot \left(1 - \hat{y}_{ij}\right)^\gamma \hat{y}_{ij} \ , y_{ij} = 1 \\ q_j \cdot \hat{y}_{ij}^\gamma \left(1 - \hat{y}_{ij}\right) \ , y_{ij} = 0 \end{cases} , \tag{9}$$

where q_j is the ratio of the positive samples for the jth attribute in each batch; γ ($\gamma \geq 0$) is an adjustable parameter.

Finally, by combining the adaptive Focal loss penalty term L_{AF} with the MSE loss L_{MSE}, the total adaptive Focal loss function AFL used in our method can be written as

$$AFL = L_{MSE} + \beta \frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{M} L_{AF} \ . \tag{10}$$

Here, β is a tuning hyperparameter.

4. Experiments

In this section, we systematically conduct experiments to test the effectiveness of the proposed APS method for FAR. In Section 4.1, we describe two public FAR datasets and parameter settings. In Section 4.2, we perform an ablation study to thoroughly evaluate the effect of the main components of the proposed method. In Section 4.3, we compare the performance of the proposed method with several state-of-the-art FAR methods.

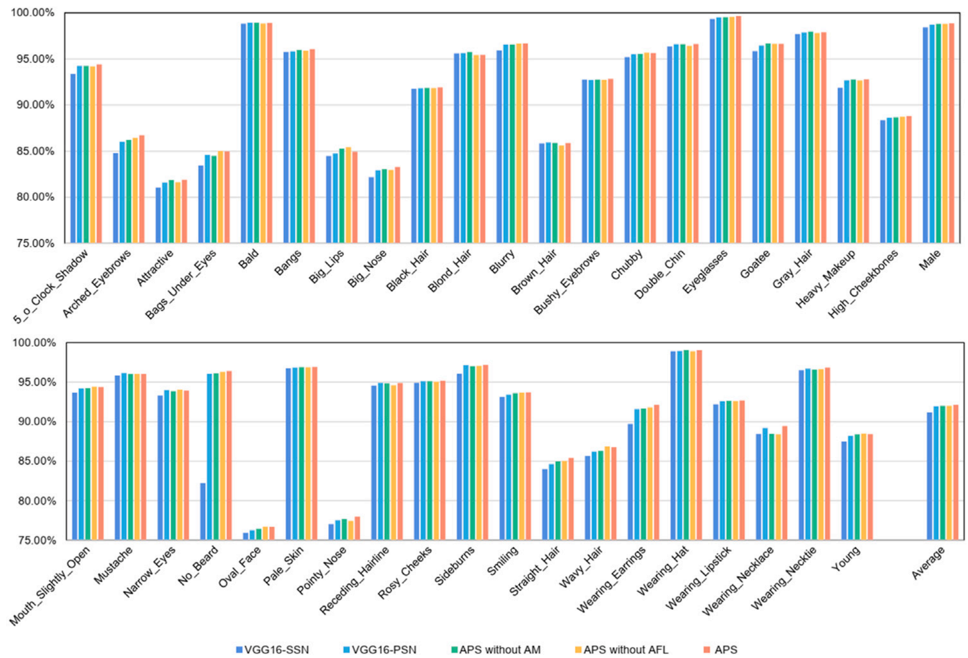

Fig. 3. Performance comparisons of different attributes among the different variants of the proposed APS method on the CelebA dataset.

4.1. Datasets and parameter settings

Due to personal privacy and the difficulty in labeling, only two public datasets, i.e., CelebA and LFWA, are widely used in the existing FAR works [6,19,23,47–49]. The CelebA [6] is a widely-used large-scale facial attribute dataset to evaluate the performance of FAR. It contains more than 10K celebrity identities and 200K facial images, each with annotations of 40 binary attributes and five landmark locations. The CelebA not only covers the different challenging factors, such as pose variation, background clutter and image quality, but also has an imbalanced class distribution. As in [6], we use the aligned CelebA dataset and split it into 19 962 test images, 19 962 validation images, and 162 770 training images.

The LFWA [6] is another public facial attribute dataset, which is originally collected based on the face recognition dataset LFW, and then added with binary attribute labels. LFWA contains 13 232 images of 5749 identities and the same 40 attribute annotations as in the CelebA dataset. As in [6], this dataset is divided into two approximately equal partitions for the training and test sets.

For the CelebA dataset, the input image size is resized to be 224×224, the total number of iterations to update the parameters is 15 epochs, and the batch size is 16. For the LFWA dataset, the input image size is 64×64, the total number of iterations to update the parameters is 15 epochs, and the batch size is 16. We adopt the stochastic gradient descent (SGD) algorithm with the learning rate 0.001, and the linear decay rate is set to be 10. We empirically set the hyperparameters τ, η, γ and β in Eqs. (4), (7), (9) and (10) as follows. As for τ and η, we empirically set both τ and η to be 0.1 for the best performance. Based on previous work [46], the focusing parameter γ smoothly adjusts the estimated probability to down-weigh easy examples, and we found $\gamma = 2$ to work best in our experiments. Besides, we set $\beta = 0.25$ to help the APS method achieve a balance between the MSE loss of attribute classification and the adaptive Focal loss penalty term. Note that the attribute classification is the primary task, so the weight of the MSE loss is always set as 1. We train the proposed APS method on the Pytorch platform and conduct all experiments on the NVIDIA Titan X GPU.

Table 1
The comparisons between the four variants and the proposed APS.

Variants	SSN	PSN	AM	AFL
VGG16-SSN	✓			
VGG16-PSN		✓		
APS without AM		✓		✓
APS without AFL		✓	✓	
APS		✓	✓	✓

4.2. Ablation study

4.2.1. Analysis of different components

In the experiments, we perform the adequate ablation studies over different components of the proposed network to explore their contributions. The proposed APS combines three main components, i.e., parallel sharing network (PSN), attention mechanism with multi-feature soft-alignment modules (AM), and adaptive Focal loss function (AFL). We evaluate the different variants of the proposed APS method, and the details of the different variants are listed in Table 1. Specifically, we use VGG-16 as baseline with the standard multi-task learning, which splits at the last layer for the final prediction of each attribute. The serial sharing network (SSN) is used in the variant VGG16-SSN. In addition, the parallel sharing network (PSN) is used in the variant VGG16-PSN, while the attention mechanism is deleted from the proposed method in the variant APS without AM. The traditional MSE loss is adapted as the loss function in the variant APS without AFL.

The ablation results are shown in Figs. 3 and 4, and we have several key observations as follows. The comparisons between VGG16-SSN and VGG16-PSN show that the parallel sharing network achieves better performance than the serial sharing network (especially on the "Pointy Nose", "Oval Face"and "Young" attributes), which can prove the effectiveness of the parallel sharing network to perform multi-task learning. Moreover, APS achieves the higher classification accuracy compared with APS without AM in terms of the recognition rate on the most attributes. Specifically, the average recognition rate of APS is higher than APS without AM by 0.14% and 0.61% improvements on CelebA and LFWA, respectively. Therefore, we can see that the

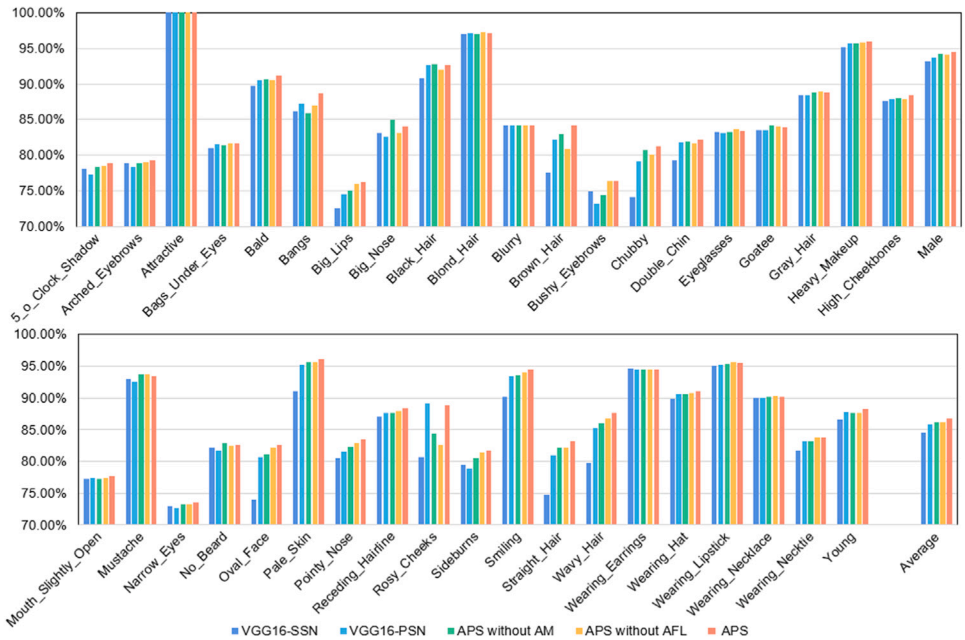

Fig. 4. Performance comparison among the different variants of the proposed APS method on the LFWA dataset.

attention mechanism is beneficial to improve the performance of FAR by exploiting the intrinsic relationship between the global and local features obtained from the shared sub-network and the two task-specific sub-networks. In addition, by comparing APS without AFL and the proposed APS method, we can see that APS achieves better accuracy. Specifically, APS without AFL achieves 92.00% (86.20%) accuracy, while APS obtains 92.12% (86.74%) accuracy on CelebA (LFWA), which shows the effectiveness of the adaptive Focal loss penalty scheme for improving the recognition performance. Generally, APS obtains the best accuracy among all variants (i.e., 92.12% and 86.74% average accuracy on CelebA and LFWA, respectively), which can be attributed to the proposed parallel sharing network, the attention mechanism and the adaptive Focal loss penalty scheme. In addition, from the average values of Figs. 3 and 4, we can observe that the parallel sharing network has the greatest influence on the FAR performance among the three main components on the CelebA and LFWA datasets. This is because the parallel sharing network can automatically transfer the important shared information learned from the different blocks of the shared sub-network to the two task-specific sub-networks, thus significantly improving the recognition performance of FAR.

4.2.2. Influence of the number of the multi-feature soft-alignment modules

The attention mechanism plays a crucial role in the performance of FAR. To test the influence of the number of multi-feature soft-alignment modules in the attention mechanism on FAR performance, Fig. 5 shows the average accuracy with the different numbers of modules on the CelebA and LFWA datasets, respectively. We observe that the recognition performance is the best when there are the two multi-feature soft-alignment modules in our network (i.e., on CelebA and LFWA, the accuracy values are 92.12% and 86.74%, respectively). The reasons for this phenomenon are probably that, the global feature $\mathbf{g}^t$ is the output of the linear classification layer at the end of the network, and the attention value of each multi-feature soft-alignment module is evaluated by a compatibility score between the local features and the global feature. The compatibility score of the local features should be higher when they are obtained from the deeper network levels, because the features from the deeper network are more beneficial for attribute classification. Thus we employ the two multi-feature soft-alignment modules in the deeper levels of the network to obtain higher compatibility scores for better recognition performance. The results of Fig. 5 have demonstrated the proposed attention mechanism can significantly improve the performance. Therefore, we set the number of multi-feature soft-alignment modules to be 2 and these two modules are connected with the outputs of the last two blocks for the shared and task-specific sub-networks in our experiments.

Table 2
Experimental results of different compatibility schemes on the CelebA and LFWA datasets (the best results are marked in bold).

Compatibility scheme		Mean accuracy (%)	
		CelebA	LFWA
c_i^{tb}	(see Eq. (2))	92.08	85.72
r_i^{b}	(see Eq. (3))	91.95	84.49
$\hat{c}_i^{tb}$	(see Eq. (4))	**92.12**	**86.74**

4.2.3. Analysis of different compatibility schemes

For each multi-feature soft-alignment module, we further evaluate the effectiveness of the following three compatibility schemes: (1) Only using c_i^{tb} in Eq. (2) to combine the local features from the task-specific sub-networks with the global feature $\mathbf{g}^t$; (2) Only using r_i^b in Eq. (3) to combine the local features from the shared sub-networks with the global feature $\mathbf{g}^t$; (3) Using $\hat{c}_i^{tb}$ in Eq. (4) to combine the local features from both the task-specific sub-networks and the shared sub-network with the global feature $\mathbf{g}^t$. Table 2 gives the experimental results of the different compatibility schemes on the CelebA and LFWA datasets. We can see that the proposed compatibility scheme can achieve the best performance, which not only considers the local and global features from the task-specific sub-networks but also fuses the local features of the shared sub-network. Therefore, it is proved that the multi-feature soft-alignment module is effective to take advantage of the relationships between the local and global features as well as between the shared sub-network and the task-specific sub-networks.

4.3. Visualizing network response using class activation maps

To intuitively comprehend the capability of the proposed method in separating the attribute classes, we show the class activation maps (CAM) [50] of APS to visualize the attention maps of the network on the CelebA dataset, as shown in Fig. 6. Obviously, these class

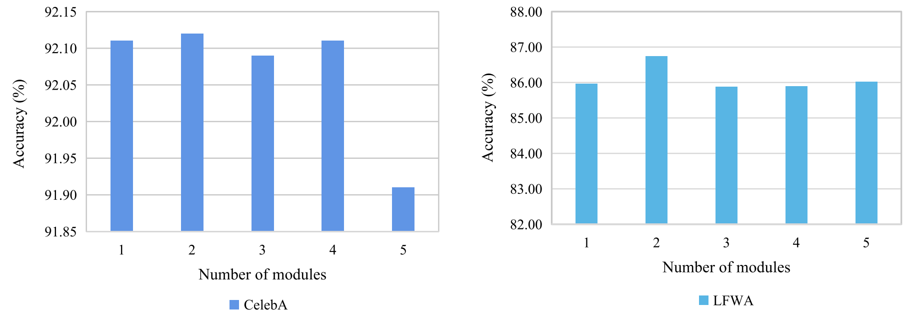

Fig. 5. Accuracy of the proposed APS method with the different numbers of the multi-feature soft-alignment modules on CelebA and LFWA.

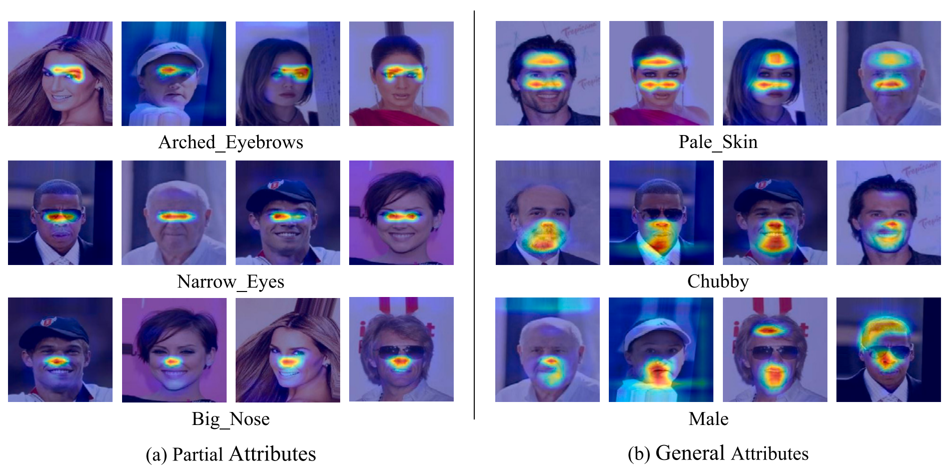

Fig. 6. Visualization of class activation maps for multiple facial attributes in the two different attribute groups.

activation maps highlight the specific locations in interpretably meaningful regions. We can see from Fig. 6 that the network focuses on the region near the top of the eyes when predicting the attribute “Arched_Eyebrows”. Similarly, the network focuses its attention on the eyes and the nose to predict the attribute “Narrow_Eyes” and “Big_Nose”, respectively. On the other hand, the high response facial regions of the general attributes is relatively larger and more irregular than those of the partial attributes. For example, for the attributes “Chubby” and “Male”, the network has the high response on approximately the half of face. Hence, these attention maps can identify the regions that are important to learn the network, and mask out uninformative regions. Meanwhile, the class activation maps in Fig. 6 demonstrate that the proposed attention-aware parallel sharing network is effective for both the partial and general attributes.

4.4. Comparison with state-of-the-art FAR methods

To further verify the performance of FAR, we compare the classification accuracy of our method with seven state-of-the-art FAR methods in Tables 3 and 4 on the CelebA and LFWA datasets, respectively. Representative FAR methods include LNets+ANet [6], MCNN-AUX [23], AFFAIR [47], MCFA [48], SPLITFACE [49], DMM-CNN [19] and SSPL [36]. Specifically, LNets+ANet [6] uses two localization networks to automatically detect the facial region and one attribute network to learn the attribute features from the detected region accordingly. MCNN-AUX [23] utilizes a multi-task deep CNN, which shares the features of the low-level layers and divides all attributes into 9 groups to classify at high-level layers. AFFAIR [47] utilizes an end-to-end network architecture to predict facial attributes without facial landmark detection and face alignment. MCFA [48] also introduces a multi-task deep CNN to jointly train multiple tasks, and these tasks include face detection, facial landmark detection and attribute recognition. SPLITFACE [49] designs the segment-based method to perform FAR for partially occluded faces. DMM-CNN [19] uses the different levels of the spatial pyramid pooling (SPP) layer to extract features for objective attributes and subjective attributes. SSPL [36] designs the spatial-semantic patch learning from large-scale unlabeled facial data to effectively perform FAR with limited labeled data.

As shown in Tables 3 and 4, the proposed APS method outperforms all the competitive methods and attains the average accuracy of 92.12% (86.74%) on CelebA (LFWA). Tables 3 and 4 show that the APS not only performs the best under average accuracy, but also obtains the best accuracy on the 16 attributes of the CelebA dataset and the 19 attributes of the LFWA dataset. Especially on attributes of “Black Hair”, “Heavy Makeup”, “High Cheekbones”, “Narrow Eyes”, and “Wearing Earrings”, the APS yields the outstanding performance compared to the other methods on CelebA. Compared with the part-based FAR method LNets+ANet, APS achieves 5.71% and 2.89% improvements in accuracy on CelebA (LFWA) by taking the advantage of the end-to-end learning pipeline. The proposed APS also beats the competitive MTL-based FAR methods, i.e., MCNN-AUX, MCFA, DNN-CNN and SSPL, significantly improving the accuracy by 0.83% (0.43%), 0.89% (3.11%), 0.42% (0.18%) and 0.44% (0.40%) on CelebA (LFWA), respectively. In addition, it can be seen that APS achieves higher accuracy than MCFA, DMM-CNN and SSPL, even without the auxiliary facial related tasks. The existing MTL-based FAR methods use the serial sharing network that ignores the importance of low-level features. It is remarkable that our method takes advantage of the parallel sharing network and multi-feature soft-alignment modules, which fully exploits the important feature information from the different levels in the shared sub-network

Table 3
Recognition accuracy (%) obtained by the proposed APS and the state-of-the-art methods on the CelebA dataset, where the upper part shows the partial attributes, and the lower part shows the general attributes (the best results are marked in bold).

Attribute	LNets+ANet	MCNN-AUX	AFFAIR	MCFA	SPLITFACE	DMM-CNN	SSPL	APS
Attractive	81.00	83.06	83.21	83.00	82.76	**83.37**	82.65	81.89
Blurry	84.00	96.17	96.43	96.00	95.96	96.40	96.51	**96.68**
Chubby	91.00	95.67	95.78	96.00	95.94	95.86	**96.12**	95.63
Heavy_Makeup	90.00	91.55	91.92	92.00	91.59	91.85	92.01	**92.80**
Male	98.00	98.17	98.46	98.00	97.95	98.29	**98.86**	**98.86**
Oval_Face	66.00	75.84	76.31	75.00	74.93	75.89	**76.98**	76.70
Pale_Skin	91.00	97.05	**97.18**	97.00	97.00	97.00	97.14	96.91
Rosy_Cheeks	90.00	95.16	**95.32**	95.00	94.79	95.32	**95.68**	95.18
Smiling	92.00	92.73	93.23	93.00	92.70	93.22	93.35	**93.69**
Straight_Hair	73.00	83.58	83.46	85.00	80.41	84.72	83.78	**85.41**
Wavy_Hair	80.00	83.91	84.01	85.00	81.70	86.01	84.69	**86.76**
Young	87.00	88.48	88.44	88.00	88.01	**88.98**	88.82	88.42
5_o_Clock_Shadow	91.00	94.51	94.79	94.00	93.13	94.84	**95.22**	94.41
Arched_Eyebrows	79.00	83.42	83.99	83.00	82.56	84.57	84.32	**86.73**
Bags_Under_Eyes	79.00	84.92	85.16	85.00	84.86	**85.81**	85.32	84.96
Bald	98.00	98.90	98.93	99.00	98.03	99.03	**99.16**	98.92
Bangs	95.00	96.05	96.15	96.00	95.71	**96.22**	96.03	96.06
Big_Lips	68.00	71.47	71.84	72.00	69.28	72.93	71.43	**84.93**
Big_Nose	78.00	84.53	84.54	84.00	83.81	84.78	**85.56**	83.28
Black_Hair	88.00	89.78	90.11	89.00	89.03	90.50	89.10	**91.92**
Blond_Hair	95.00	96.01	**96.14**	96.00	95.76	96.13	96.01	95.44
Brown_Hair	80.00	89.15	89.55	88.00	88.25	89.46	**90.16**	85.87
Bushy_Eyebrows	90.00	92.84	92.89	92.00	92.66	**93.01**	92.99	92.85
Double_Chin	92.00	96.32	96.41	96.00	95.80	96.39	**97.06**	96.61
Eyeglasses	99.00	99.63	99.71	**100.00**	99.51	99.69	99.65	99.65
Goatee	95.00	97.24	97.60	97.00	96.68	97.63	**97.84**	96.63
Gray_Hair	97.00	98.20	98.24	98.00	97.45	98.27	**98.88**	97.91
High_Cheekbones	88.00	87.58	88.13	87.00	87.61	87.73	88.23	**88.82**
Mouth_Slightly_Open	92.00	93.74	94.27	93.00	93.78	94.16	94.28	**94.38**
Mustache	95.00	96.88	**97.03**	97.00	95.86	**97.03**	97.00	96.04
Narrow_Eyes	81.00	87.23	87.84	87.00	86.88	87.73	87.57	**93.92**
No_Beard	95.00	96.05	96.37	96.00	96.17	96.41	**96.89**	96.42
Pointy_Nose	72.00	77.47	77.49	77.00	76.47	77.19	77.14	**77.96**
Receding_Hairline	89.00	93.81	93.75	94.00	92.25	94.12	94.22	**94.89**
Sideburns	96.00	97.85	97.93	**98.00**	97.17	97.91	97.98	97.17
Wearing_Earrings	82.00	90.43	90.51	90.00	89.44	90.78	90.58	**92.12**
Wearing_Hat	99.00	99.05	99.12	99.00	98.74	99.12	**99.23**	99.04
Wearing_Lipstick	93.00	94.11	94.04	94.00	93.21	**94.49**	93.35	92.68
Wearing_Necklace	71.00	86.63	86.48	88.00	85.61	88.03	88.14	**89.44**
Wearing_Necktie	93.00	96.51	95.25	97.00	96.05	97.15	**97.25**	96.84
Average	87.33	91.29	91.45	91.23	90.61	91.70	91.68	**92.12**

to improve performance. In addition, the AFL can handle the problems of class imbalance and hard example mining for boosting the FAR performance.

5. Conclusion

In this paper, we propose a novel attention-aware parallel sharing network for multi-task learning based FAR. We first introduce a parallel sharing network, in which the importance of each block of the shared sub-network can be adaptively learned for the two task-specific sub-networks, in order to fully exploit the useful low-level spatial information. To further improve the classification ability of discriminating the attributes, we introduce an attention mechanism with the multi-feature soft-alignment modules to leverage the compatibility between global and local features from both the shared sub-network and the task-specific sub-networks. Besides, we develop an adaptive Focal loss penalty scheme to alleviate the problems of class imbalance and hard example mining. Experimental results on the public CelebA and LFWA datasets have demonstrated that the proposed APS method outperforms several state-of-the-art FAR methods.

CRediT authorship contribution statement

Si Chen: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Formal analysis, Resources, Software, Validation, Supervision, Project administration. **Xinyu Lai:** Methodology, Writing – original draft, Writing – review & editing, Software, Visualization, Validation, Investigation, Data curation. **Yan Yan:** Resources, Writing – review & editing, Investigation. **Da-Han Wang:** Resources, Writing – review & editing. **Shunzhi Zhu:** Investigation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

CelebA: http://mmlab.ie.cuhk.edu.hk/projects/CelebA.html; LFWA: https://talhassner.github.io/home/projects/lfwa/.

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Table 4
Recognition accuracy (%) obtained by the proposed APS and the state-of-the-art methods on the LFWA dataset, where the upper part shows the partial attributes, and the lower part shows the general attributes (the best results are marked in bold).

Attribute	LNets+ANet	MCNN-AUX	AFFAIR	MCFA	SPLITFACE	DMM-CNN	SSPL	APS
Attractive	83.00	80.31	80.31	77.00	80.16	81.10	80.64	**100.00**
Blurry	74.00	85.23	86.82	86.00	86.42	**87.58**	86.54	84.19
Chubby	73.00	76.86	75.71	74.00	76.06	77.66	77.27	**81.24**
Heavy_Makeup	95.00	95.85	95.86	94.00	95.39	95.68	95.27	**95.95**
Male	94.00	94.02	93.94	93.00	92.60	94.14	93.21	**94.55**
Oval_Face	74.00	77.39	79.32	74.00	76.80	76.94	78.59	**82.56**
Pale_Skin	84.00	93.32	91.24	82.00	90.97	91.86	91.88	**96.12**
Rosy_Cheeks	78.00	87.92	87.56	85.00	87.08	86.44	88.65	**88.79**
Smiling	91.00	91.83	91.51	88.00	90.80	92.24	90.45	**94.43**
Straight_Hair	76.00	78.53	78.58	77.00	78.91	79.20	81.06	**83.17**
Wavy_Hair	76.00	81.61	79.72	79.00	78.28	79.87	81.22	**87.60**
Young	86.00	85.84	85.93	87.00	85.68	**88.94**	88.25	88.22
5_o_Clock_Shadow	**84.00**	77.06	77.98	75.00	77.59	79.18	80.28	78.94
Arched_Eyebrows	82.00	81.78	81.63	79.00	81.72	**82.70**	81.56	79.29
Bags_Under_Eyes	83.00	**83.48**	83.12	79.00	82.62	82.70	82.19	81.70
Bald	88.00	91.94	91.67	91.00	91.88	91.96	**92.58**	91.16
Bangs	88.00	90.08	91.08	89.00	90.71	**91.30**	90.98	88.74
Big_Lips	75.00	79.24	78.91	75.00	78.97	**79.82**	77.63	76.17
Big_Nose	81.00	**84.98**	84.20	81.00	83.13	83.67	84.02	84.05
Black_Hair	90.00	92.63	91.66	91.00	92.49	91.55	91.23	**92.67**
Blond_Hair	97.00	97.41	**97.56**	97.00	97.47	97.17	97.05	97.20
Brown_Hair	77.00	80.85	79.35	77.00	80.93	81.56	82.78	**84.14**
Bushy_Eyebrows	82.00	84.97	84.90	76.00	84.26	**85.33**	84.56	76.37
Double_Chin	78.00	81.52	81.00	77.00	80.49	80.98	81.14	**82.20**
Eyeglasses	95.00	91.30	92.25	91.00	91.50	92.83	**93.05**	83.34
Goatee	78.00	82.97	83.56	80.00	83.01	82.82	83.11	**83.98**
Gray_Hair	84.00	88.93	88.14	88.00	88.46	**89.38**	89.14	88.83
High_Cheekbones	88.00	88.38	**88.85**	85.00	88.34	88.13	87.98	88.42
Mouth_Slightly_Open	82.00	83.51	81.48	78.00	82.50	**84.45**	82.06	77.78
Mustache	92.00	93.43	93.71	91.00	92.97	**94.46**	93.54	93.41
Narrow_Eyes	81.00	82.86	82.08	78.00	82.75	**83.67**	81.72	73.57
No_Beard	79.00	82.15	81.34	79.00	80.77	82.48	81.54	**82.55**
Pointy_Nose	80.00	84.14	**84.94**	80.00	84.20	84.51	84.28	83.49
Receding_Hairline	85.00	86.25	86.13	85.00	84.90	86.30	85.25	**88.42**
Sideburns	77.00	**83.13**	83.10	78.00	81.76	82.99	81.48	81.77
Wearing_Earrings	94.00	94.95	**95.07**	93.00	94.75	94.14	94.87	94.49
Wearing_Hat	88.00	90.07	89.84	91.00	90.23	90.84	**91.27**	91.07
Wearing_Lipstick	95.00	95.04	94.77	94.00	94.07	95.11	94.36	**95.48**
Wearing_Necklace	88.00	89.94	89.77	89.00	89.59	89.47	89.87	**90.11**
Wearing_Necktie	79.00	80.66	80.71	82.00	81.40	81.28	81.24	**83.78**
Average	83.85	86.31	86.13	83.63	85.82	86.56	86.34	**86.74**

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